

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 9

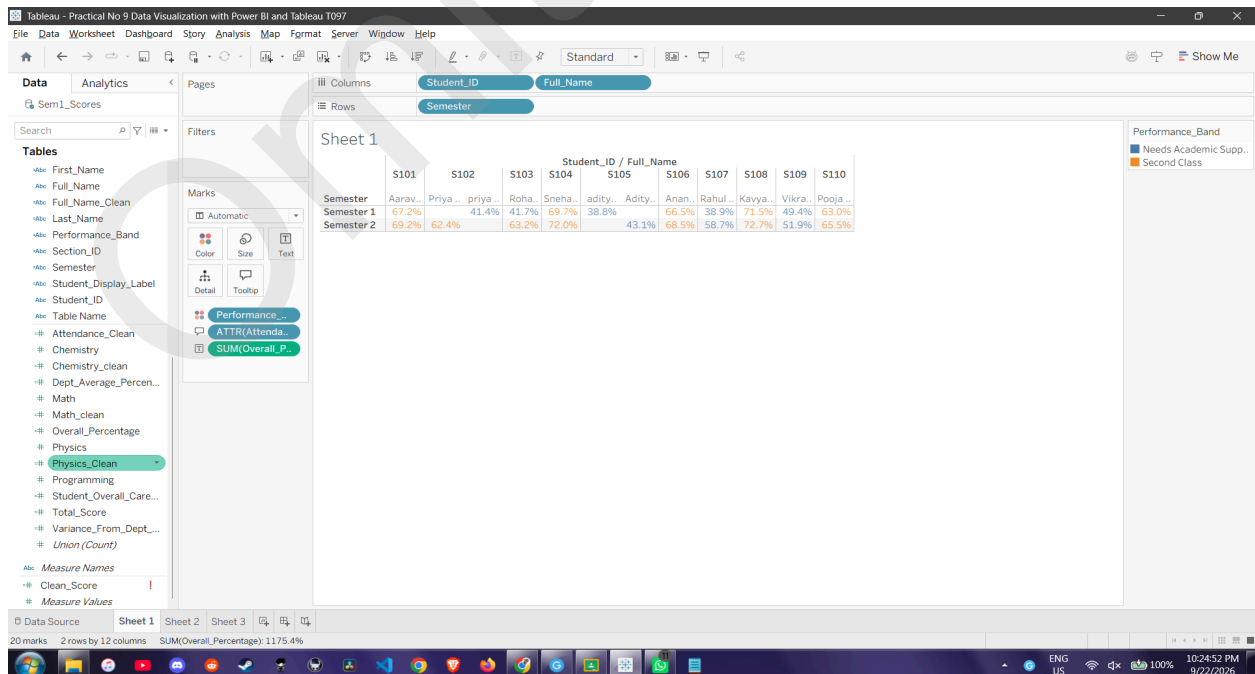
Aim:

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

a) Identify and handle missing or inconsistent data

Task 1: Clean Inconsistent String Fields

- Load **Sem1_Scores.csv** into Tableau Desktop.
- Navigate to Sheet 1. Create a Calculated Field named **Physics_Clean** with the formula:
FLOAT(REPLACE(STR([Physics]), '%', '')). Click **OK**.
- Create a Calculated Field named **Attendance_Clean** with the formula:
FLOAT(TRIM(REPLACE(STR([Attendance]), '%', ''))). Click **OK**.

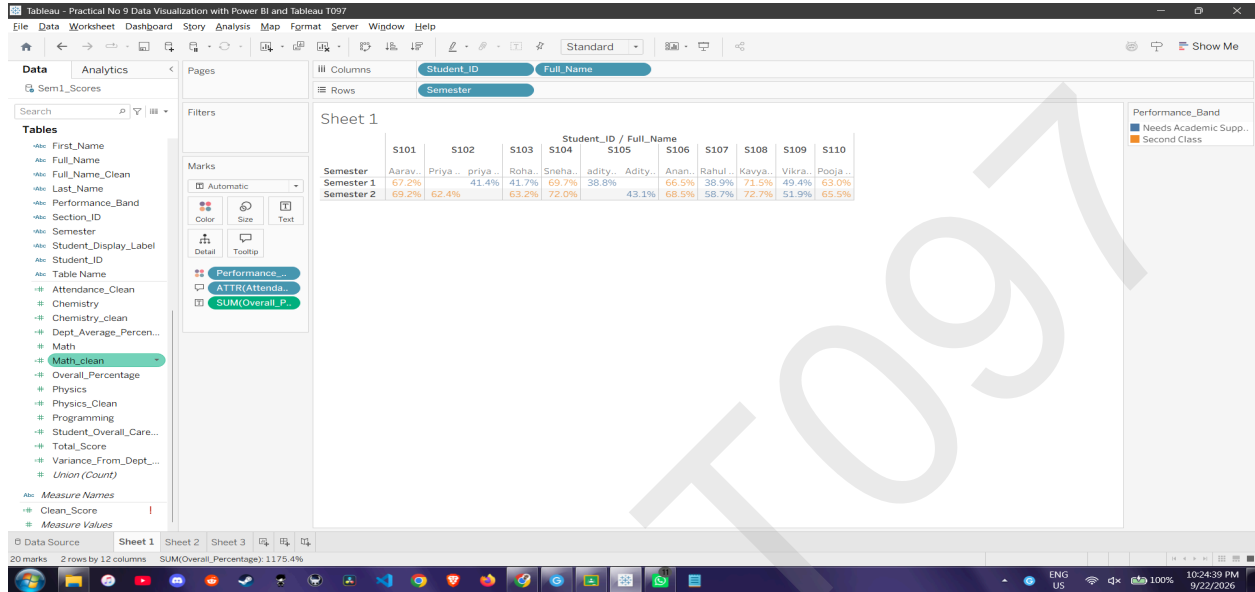


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Task 2: Handle Missing (Null) Values via Imputation

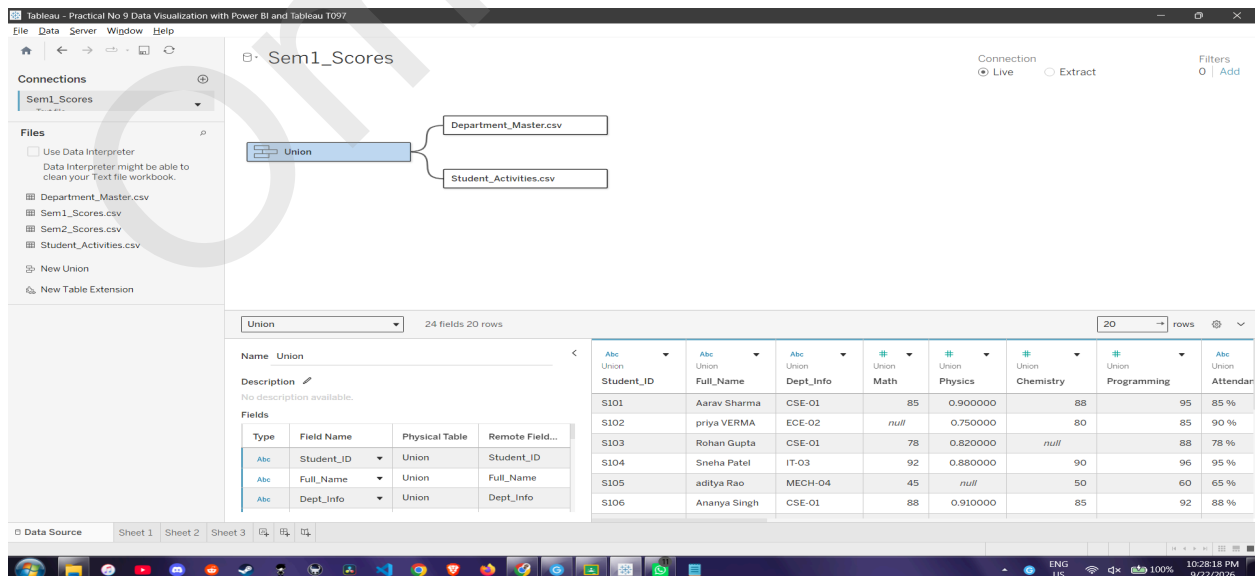
- Create a Calculated Field named **Math_Clean** with the formula: **IFNULL([Math], 0)**. Click **OK**.
- Create a Calculated Field named **Chemistry_Clean** with the formula: **IFNULL([Chemistry], 0)**. Click **OK**.



b) Convert wide data into tall format

Task 1: Pivot Subject Columns

- Navigate to the **Data Source** tab at the bottom-left corner.
- Select the **Math**, **Physics**, **Chemistry**, and **Programming** column headers by holding Ctrl / Cmd. Right-click any of the highlighted headers and select **Pivot**.



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Data Visualization with Power BI and Tableau

Task 2: Rename and Format Pivoted Fields

- Double-click the new **Pivot Field Names** header and rename it to **Subject**.
- Double-click the new **Pivot Field Values** header and rename it to **Score**.
- Click the **Abc** icon directly above the **Score** column header and select **Number (decimal)**.

Tableau - Practical No 9 Data Visualization with Power BI and Tableau T097

Connections: Sem1_Scores

Files: Department_Master.csv, Sem1_Scores.csv, Sem2_Scores.csv, Student_Activities.csv, New Union, New Table Extension

Union

24 fields 20 rows

Name: Union

Description: No description available.

Fields:

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendar
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %

c) Perform union of multiple files

Task 1: Create a Multi-File Union

- In the **Data Source** tab, clear the canvas. Drag the **New Union** option onto the empty canvas.
- Drag **Sem1_Scores.csv** and **Sem2_Scores.csv** from the left panel into the pop-up Union dialog box. Click **Apply**, then **OK**.

Tableau - Practical No 9 Data Visualization with Power BI and Tableau T097

Connections: Sem1_Scores

Files: Department_Master.csv, Sem1_Scores.csv, Sem2_Scores.csv, Student_Activities.csv, New Union, New Table Extension

Union

24 fields 20 rows

Name: Union

Description: No description available.

Fields:

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info

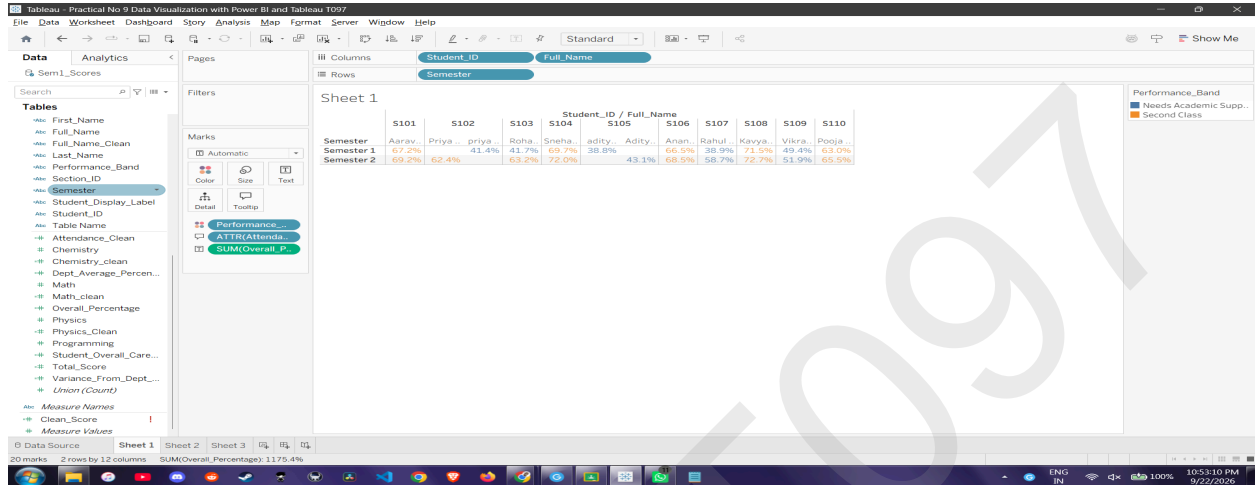
Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendar
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	75	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %

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Task 2: Create a Clean Semester Identifier

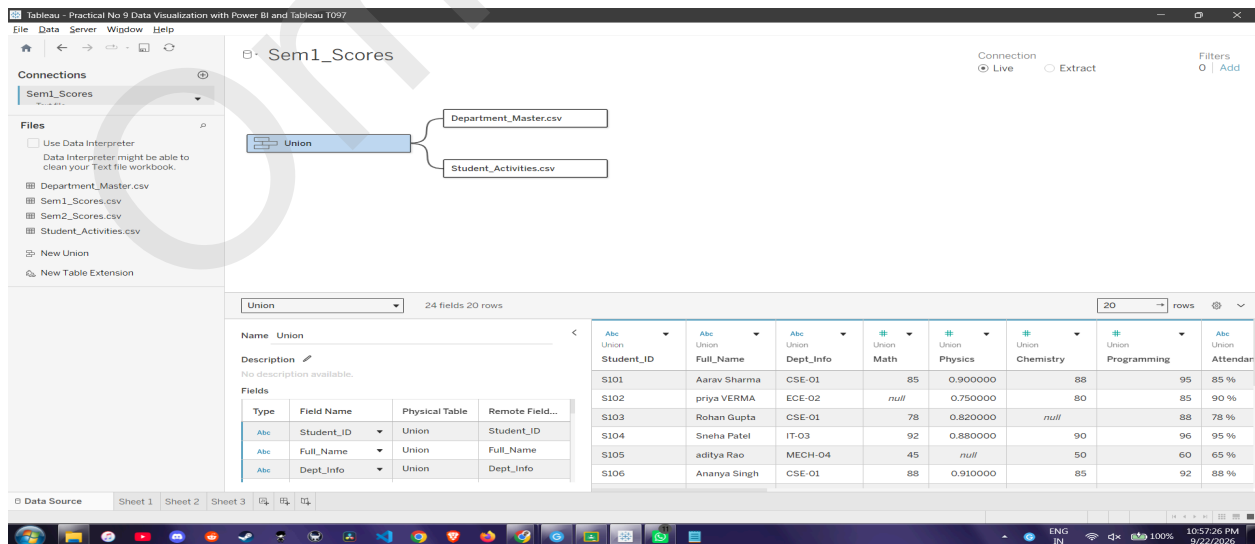
- Navigate to a worksheet. Create a Calculated Field named **Semester**.
- Formula: **IF CONTAINS([Table Name], "Sem1") THEN "Semester 1" ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2" ELSE "Other" END**. Click **OK**.



d) Execute cross-database joins

Task 1: Link Department and Activity Data

- In the **Data Source** tab, drag **Department_Master.csv** onto the canvas and drop it to the right of your Union box.
- In the relationship dialog, set Left Table Field to **Dept_Info** and Right Table Field to **Dept_Code**.
- Drag **Student_Activities.csv** onto the canvas to the right of the Union box. Set the relationship to match **Student_ID = Student_ID**.



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Data Visualization with Power BI and Tableau

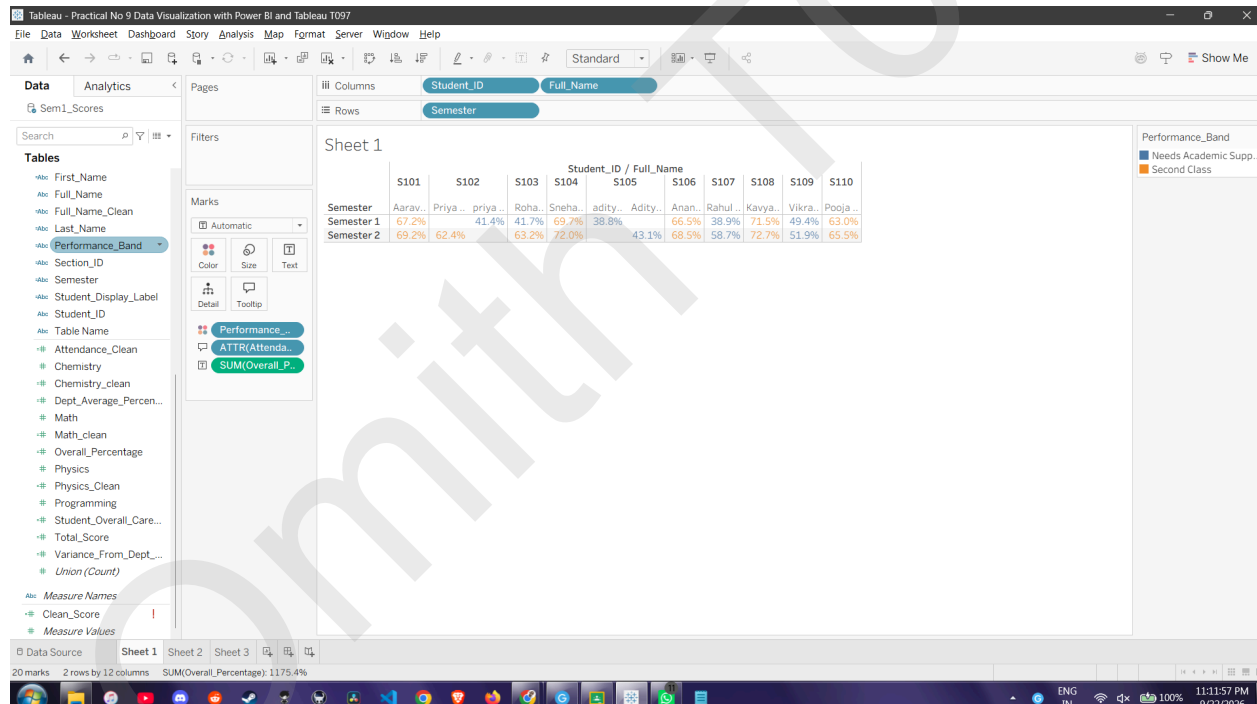
e) Restructure data for better visualization

Task 1: Calculate Total Marks and Scaled Percentage

- Create a Calculated Field named **Total Score** with the formula: $[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]$.
- Create a Calculated Field named **Overall_Percentage** with the formula: $[Total_Score] / 400.0$.
- Right-click **Overall_Percentage** $\rightarrow$ **Default Properties** $\rightarrow$ **Number Format** $\rightarrow$ **Percentage** (1 decimal place).

Task 2: Create Categorical Performance Bands

- Create a Calculated Field named **Performance_Band**. Formula: IF $[Overall_Percentage] \geq 0.85$ THEN "Distinction" ELSEIF $[Overall_Percentage] \geq 0.75$ THEN "First Class" ELSEIF $[Overall_Percentage] \geq 0.60$ THEN "Second Class" ELSE "Needs Academic Support" END.
- Create a Calculated Field named **Attendance_Status**. Formula: IF $[Attendance_Clean] \geq 75$ THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END.



SHETH L.U.J & SIR M.V COLLEGE
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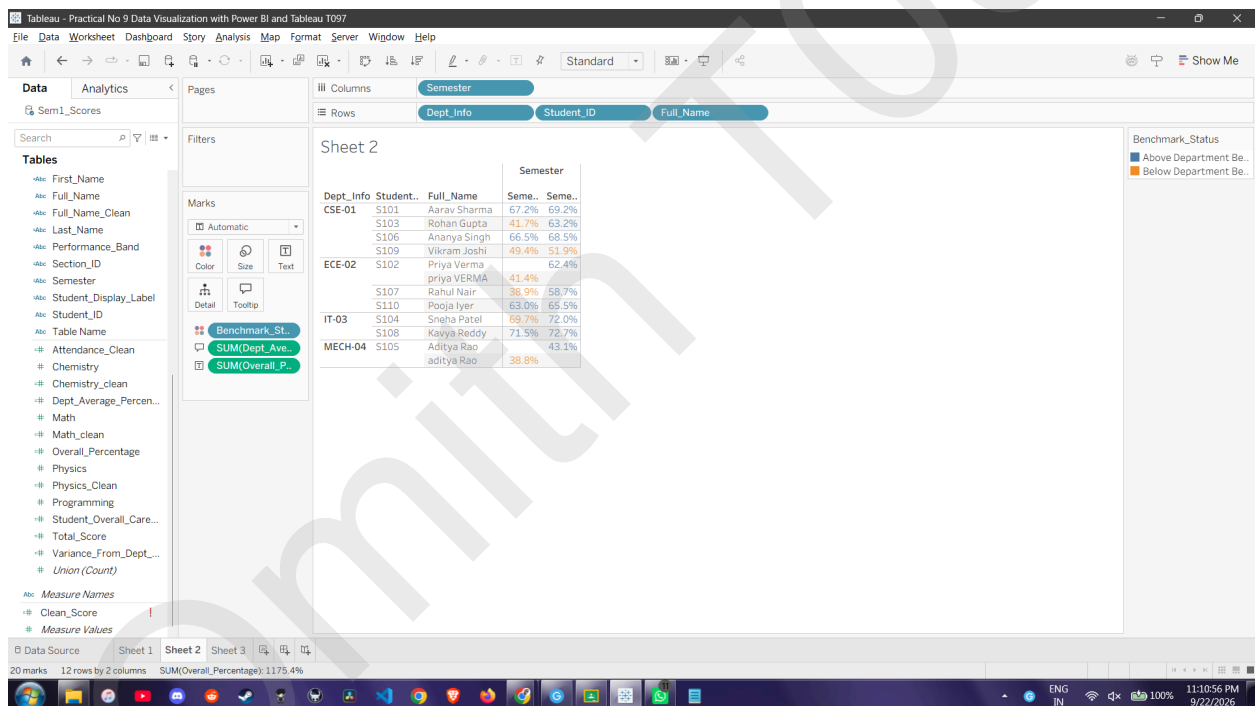
f) Handle different levels of detail in datasets

Task 1: Compute Benchmark LODs

- Create a Calculated Field named **Dept_Average_Percentage_LOD**. Formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }.
- Create a Calculated Field named **Student_Overall_Career_Average**. Formula: { FIXED [Student_ID]: AVG([Overall_Percentage]) }.

Task 2: Calculate Performance Variance

- Create a Calculated Field named **Variance From Dept Average**. Formula: [Overall_Percentage] - [Dept_Average_Percentage_LOD].
- Create a Calculated Field named **Benchmark_Status**. Formula: IF [Variance From Dept Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END.



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Data Visualization with Power BI and Tableau

g) Apply advanced data cleaning techniques

Task 1: Clean Whitespace and Parse Names

- Create a Calculated Field named **Full_Name_Clean**. Formula: `TRIM([Full_Name])`.
- Create a Calculated Field named **First_Name**. Formula: `TRIM(SPLIT([Full_Name], " ", 1))`.
- Create a Calculated Field named **Last_Name**. Formula: `TRIM(SPLIT([Full_Name], " ", -1))`.

Task 2: Split Composite Codes and Combine Labels

- Create a Calculated Field named **Dept_Branch**. Formula: `SPLIT([Dept_Info], "-", 1)`.
- Create a Calculated Field named **Section_ID**. Formula: `SPLIT([Dept_Info], "-", 2)`.
- Create a Calculated Field named **Student_Display_Label**. Formula: `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"`.

